

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Data Driven Booking and Occupancy Analytics System for Apartment Management

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The rapid advancement of digital technology has significantly transformed the way businesses operate, particularly in the accommodation and hospitality industry. Traditional methods of handling reservations—such as walk-ins, phone calls, and manual logbooks—have gradually been replaced by web-based and mobile booking systems. These digital platforms provide greater convenience, accessibility, and efficiency for both customers and business owners. As a result, many accommodation providers now utilize online platforms to manage bookings and communicate with clients.

With the emergence of platforms such as Airbnb and social media channels like Facebook, customers are now able to inquire about and reserve rooms with just a few clicks. These platforms have contributed to increased visibility for small lodging businesses, allowing them to reach a broader audience without the need for extensive marketing resources. However, while these tools provide convenience, they also introduce new challenges, especially for small-scale accommodation providers that lack integrated management systems.

Donadel Hometel, a small lodging business offering both transient and long-term accommodations, currently relies on multiple communication channels to receive booking inquiries. These include messaging platforms, social media, and direct communication with customers. While this approach enables flexibility, it also creates a fragmented system where reservation information is scattered across different platforms. As the number of inquiries increases, it becomes increasingly difficult for staff to monitor, track, and respond to all customer messages in a timely and organized manner.

One of the major challenges associated with this setup is the risk of missed or delayed responses. When inquiries are overlooked or not addressed promptly, potential customers may choose alternative accommodations, leading to lost business opportunities. Additionally, without a centralized booking system, it becomes difficult to ensure that all confirmed reservations are properly recorded and updated. This often results in inconsistencies in booking records and increases the likelihood of human error.

Another critical issue faced by the business is overbooking. Overbooking occurs when multiple reservations are accepted for the same room and time period due to the lack of synchronized scheduling and real-time updates. This problem not only affects operational efficiency but also has serious implications for customer satisfaction. Guests who experience booking conflicts may lose trust in the service, resulting in negative reviews and reputational damage. Furthermore, platforms like Airbnb may impose penalties on hosts who fail to manage bookings properly, including cancellations, refunds, and temporary account restrictions.

In addition to operational challenges, the current system lacks the capability to generate meaningful insights from booking data. Reservation information is typically recorded manually or stored in separate tools, making it difficult to analyze trends such as peak booking periods, occupancy rates, and customer preferences. Without access to these insights, business owners are limited in their ability to make informed decisions regarding pricing strategies, marketing efforts, and resource allocation.

The growing importance of data-driven decision-making in modern business environments highlights the need for systems that not only manage operations but also provide analytical capabilities. Data-driven systems enable organizations to collect, process, and interpret data in order to identify patterns, predict outcomes, and improve overall performance. In the context of accommodation management, such systems can provide valuable information on booking behavior, seasonal demand, and occupancy trends.

In response to these challenges, there is a clear need for an integrated solution that consolidates booking management processes into a single platform. A web-based reservation and occupancy analytics system can address the limitations of manual and

fragmented approaches by providing real-time tracking, automated scheduling, and centralized data storage. By streamlining operations and reducing the risk of errors, such a system can enhance efficiency and improve customer satisfaction.

This study proposes the development of a Data-Driven Booking and Occupancy Analytics System specifically designed for Donadel Hometel. The proposed system aims to provide a comprehensive solution that supports reservation management, prevents overbooking, and generates analytical reports based on booking data. Through the integration of web-based technologies and data analytics, the system seeks to improve operational processes and support informed decision-making.

1.2 Statement of the Problem

Despite the increasing use of digital communication platforms for accommodation reservation, Donadel Hometel continues to experience difficulties in effectively managing booking operations due to the absence of a centralized reservation system. The use of manual processes and fragmented communication channels creates operational inefficiencies that may affect customer satisfaction and business performance.

Specifically, this study seeks to address the following problems:

- 1.**How can the system improve the monitoring and management of reservation inquiries from multiple sources?
- 2.**How can the proposed system help reduce overlooked or delayed responses to customer reservations?
- 3.** How can automated calendar management help prevent overbooking and scheduling conflicts?

4. How can the system minimize operational issues, customer dissatisfaction, and revenue loss caused by reservation errors
5. How can the proposed system improve the accuracy and efficiency of reservation record management?
6. How can the system generate useful analytics and reports regarding booking trends, occupancy rates, and customer behavior?

1.3 Objectives of the Study

1.3.1 General Objective

This study aims to design and develop a data-driven web-based booking and occupancy analytics system for Donadel Hometel that will improve reservation management, reduce overbooking incidents, and provide analytical insights to support efficient business decision-making.

1.3.2 Specific Objectives

In particular, the study will:

1. Analyze the existing reservation process of Donadel Hometel and identify the challenges encountered in managing bookings and customer inquiries.
2. Design a user-friendly web-based reservation system that allows customers to book rooms and enables administrators to manage reservation data efficiently.
3. Develop a centralized reservation management feature that improves the tracking, monitoring, and updating of booking records.
4. Implement an automated calendar management feature to minimize overbooking and scheduling conflicts.

5. Generate occupancy reports, booking trends, and customer-related analytics to support data-driven decision-making.
6. Evaluate the system in terms of functionality, usability, efficiency, reliability, and accuracy.

1.4 Scope and Delimitation

This study focuses on the design and development of a data-driven web-based booking and occupancy analytics system for Donadel Hometel. The proposed system is intended to improve the management of reservations by providing a centralized platform where customers can book rooms online and administrators can efficiently monitor and manage reservation records.

The system will include features such as online reservation booking, reservation tracking, calendar management, occupancy monitoring, and report generation. Customers will be able to submit booking requests through the website, while administrators will have access to tools for confirming reservations, updating booking schedules, and monitoring room availability. The calendar management feature will help minimize overbooking by automatically displaying reserved and available schedules.

In addition, the system will generate reports and visual representations related to occupancy rates, booking trends, and customer reservation activities. These analytics are intended to assist management in understanding booking patterns and improving operational decision-making. The system will also maintain a database of reservation records for easier retrieval, organization, and monitoring of information.

The evaluation of the system will focus on its functionality, usability, efficiency, reliability, and accuracy based on user feedback gathered from the owner and selected staff members of Donadel Hometel.

1.5 Significance of the Study

This study is significant because it aims to address the common challenges encountered in reservation and occupancy management among small lodging businesses. Through the development of a data-driven web-based booking and occupancy analytics system, the study seeks to improve reservation handling, reduce operational inefficiencies, and support informed decision-making.

- **Business Owner / Management:**

The proposed system will benefit the owner and management of Donadel Hometel by providing a centralized and organized platform for handling reservations. The system can help reduce errors associated with manual booking processes and improve monitoring of room availability. Additionally, the analytics and reports generated by the system can support management in identifying occupancy trends, monitoring booking performance, and making better operational decisions.

- **Reservation Staff / Personnel:**

The system can help staff members manage reservations more efficiently by reducing repetitive manual tasks and simplifying reservation tracking. Through an organized interface and automated calendar management, staff can quickly monitor bookings, update schedules, and minimize the risk of overlooking customer reservations.

- **Customers / Clients:**

Customers may benefit from a more convenient and accessible reservation process through the web-based booking platform. The system can provide faster reservation processing, better schedule accuracy, and reduced risks of booking conflicts, resulting in an improved customer experience.

- **Researchers / Future Developers:**

This study may serve as a useful reference for future researchers and developers interested in developing reservation management systems, occupancy analytics tools, or web-based business solutions. The findings and system design may contribute to future studies related to booking automation, hospitality management, and data-driven systems.

- **Other Lodging Businesses:**

The proposed system may also serve as a model for other small accommodation providers seeking to improve their reservation management processes. By adopting similar technologies, lodging businesses may improve operational efficiency, enhance customer service, and reduce booking-related issues.

1.6 Definition of Terms

- **Booking / Reservation** – Refers to the process of reserving a room or accommodation unit through the system for a specified date and duration.
- **Web-Based System** – A software application that operates through a web browser and internet connection, allowing users to access booking and management functions online.
- **User** – Any individual who interacts with the system, including customers making reservations and administrators managing booking records.
- **Administrator** – An authorized system user responsible for managing reservations, monitoring booking records, generating reports, and maintaining system operations.
- **Reservation Management** – The process of organizing, monitoring, updating, and handling reservation records within the system to ensure accurate booking transactions.
- **Calendar Management** – A system feature that displays booking schedules and room availability to help prevent scheduling conflicts and overbooking.
- **Overbooking** – A situation in which multiple reservations are accepted beyond the available number of rooms or accommodation units for a specific schedule.
- **Occupancy Rate** – The percentage of occupied rooms or units within a given period in relation to the total available accommodations.

- **Booking Trends** – Patterns or behaviors observed from reservation data over time, including peak booking periods, seasonal demand, and customer reservation activities.
- **Data-Driven System** – A system that collects, processes, and analyzes data to generate useful insights and support decision-making processes.
- **Analytics** – The process of examining and interpreting reservation and occupancy data to identify trends, patterns, and useful business information.
- **Report Generation** – The process of producing summarized information such as tables, charts, and graphs based on reservation and occupancy records.
- **User-Friendly Interface** – A system design that is simple, organized, and easy to navigate for both customers and administrators.
- **Manual Booking** – A reservation process in which booking details are manually entered into the system by staff or administrators rather than automatically generated from external platforms.
- **Occupancy Analytics** - The analysis of room usage and reservation patterns to help management evaluate occupancy performance and improve decision-making.
- **Reservation Tracking** - A feature that enables administrators to monitor the status and details of reservations in real time.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Mandal, Goswami, and Kumari (2021) looked into automated systems for managing bookings. Their study showed that using automation makes things run smoother, cuts down on scheduling problems, and helps with accuracy when dealing with lots of booking data. Similarly, Fox (2020) observed that modern booking technology improves hotel reservation systems by reducing errors made by people and speeding up how fast transactions happen.

Mason and Roberts (2019) mentioned that confusing and slow booking systems make customers feel uncertain. On the other hand, reliable and easy-to-use systems create trust and increase customer happiness. In addition, Lam and Nguyen (2022) found that analyzing booking data helps to uncover patterns in how rooms are occupied and what customers like, which helps in making better choices. Ahmad and Sharma (2021) also concluded that how good a system is and how easy it is to use greatly affects customer satisfaction and their willingness to make a booking.

2.2 Local Studies

Castillo et al. (2014) stated that computerized reservation systems make hotel management easier by improving how reservations are handled and reducing mistakes. Their results highlighted the important role of automation in ensuring that bookings are accurate and in keeping track of room availability. Similarly, Delizo and Esguerra (2011) pointed out that online booking platforms have user-friendly features that help to make the booking process better and assist staff with managing reservations effectively.

Talagsad et al. (2020) found that online booking systems simplify the booking process, reduce how long it takes to process reservations, and make it easier for users to access the system. Leona et al. (2023) further demonstrated that web-based systems are popular and work well for managing reservations in local hotels. Furthermore, Cabiles et al. (2024) noted that how easy a system is to use, its reliability, and the overall user experience have a big impact on customer satisfaction and how well the operations run.

2.3 Related Literature

Data-driven systems use the information they collect to provide insights and help with decision-making (Alpaydin, 2020). For developing these systems, using strict software engineering methods is vital to ensure quality and reliability (Pressman, 2019).

Research on websites shows that how users feel about their experience, how much they trust the site, and how well the system works are really important for keeping users happy (Finkel et al., 2018). Also, online booking systems make it quicker to make reservations, make things easier for users, and reduce the amount of work businesses have to do by hand (Westchester County Business Journal, 2020).

2.4 Synthesis of the Review Literature

The reviewed foreign and local studies emphasized the growing importance of online reservation systems in improving operational efficiency, customer satisfaction, and reservation accuracy. Several studies highlighted that web-based booking systems help reduce manual errors, improve accessibility, and streamline reservation management processes.

The literature also revealed that data analytics plays a significant role in understanding occupancy trends, customer behavior, and booking patterns. Researchers emphasized that data-driven systems support better decision-making, improve booking strategies, and enhance overall business performance.

Although many existing studies focused on reservation automation and customer satisfaction, only a limited number of systems integrated booking management with occupancy analytics and reporting features. Most existing systems primarily focus on reservation processing without providing detailed occupancy analysis and decision-support capabilities.

Therefore, this study aims to address these gaps by developing a Data-Driven Booking and Occupancy Analytics System for Donadel Hometel. The proposed system integrates reservation management, calendar monitoring, occupancy tracking, and analytical reporting to improve booking efficiency and support data-driven decision-making.

CHAPTER 3

METHODOLOGY

This chapter presents the research methodology used in the development of the proposed Data-Driven Booking and Occupancy Analytics System for Donadel Hometel. It discusses the research design, system development

methodology, system architecture, tools and technologies used, data gathering techniques, testing procedures, and evaluation criteria applied in the study.

The methodology serves as a guide in designing and developing the proposed system to address the challenges encountered in reservation management, booking monitoring, and occupancy tracking. Through a systematic approach, the researchers aim to ensure that the developed system is functional, reliable, efficient, and suitable for the operational needs of Donadel Hometel.

3.1 Research Design

This study utilizes a descriptive research design to analyze the current reservation management practices of Donadel Hometel and identify the common challenges experienced in handling bookings and occupancy monitoring. Descriptive research is appropriate for this study because it allows the researchers to gather information regarding existing processes, operational issues, and user requirements related to reservation management.

The descriptive method enables the researchers to examine how reservations are currently processed, monitored, and recorded within the business. Through this approach, the researchers are able to identify problems such as missed reservations, delayed responses, scheduling conflicts, overbooking, and inefficiencies caused by manual record-keeping.

In addition, the descriptive research design helps determine the specific needs of the business and provides a basis for designing the features and functionalities of the proposed system. The information gathered from respondents contributes to the development of a system that is aligned with the operational requirements of Donadel Hometel.

However, the study may also encounter certain limitations, including limited respondents, possible response bias, and dependency on the accuracy of the information provided during interviews and surveys. Despite these limitations, the descriptive approach remains suitable for evaluating reservation

management processes and supporting the development of the proposed system.

3.1 System Development Methodology

This study adopts the Agile Software Development Methodology in developing the proposed system. Agile is a software development approach that focuses on iterative development, continuous improvement, collaboration, and regular feedback from users. This methodology is appropriate for the study because it allows the researchers to gradually develop and improve the system based on the needs and suggestions of the client.

The Agile methodology promotes flexibility and enables developers to modify system features during the development process whenever necessary. Through continuous communication with the owner and staff of Donadel Hometel, the researchers can identify issues early and make necessary adjustments to improve the system.

The development process includes these stages:

1. **Planning and Requirement Gathering** – The researchers identified the existing problems encountered by Donadel Hometel in handling reservations and occupancy monitoring. Information regarding the current booking process, reservation handling, and scheduling practices was collected through interviews, surveys, and observations. The researchers also identified the functional and non-functional requirements needed for the proposed system. Functional requirements include reservation management, booking tracking, calendar monitoring, and report generation, while non-functional requirements include usability, efficiency, and reliability.
2. **Design and Prototyping** – The researchers designed the structure and appearance of the proposed system. This includes creating the user interface design, navigation flow, database structure, and system architecture. Prototype layouts and interface designs were prepared to visualize how the system would

function. The researchers also planned the organization of reservation data, occupancy records, and reporting features to ensure efficient system performance.

3. **Incremental Development** – The proposed system was developed gradually through small and manageable phases called sprints. Each sprint focused on developing specific system functionalities such as reservation booking, reservation tracking, calendar management, occupancy monitoring, and analytics generation. This approach allowed the researchers to test individual features continuously and apply necessary improvements before proceeding to the next development phase.
4. **Testing and Feedback** – Each development phase, the researchers conducted testing procedures to evaluate the functionality and performance of the system. Unit testing, system testing, and user acceptance testing were performed to identify system errors and ensure that the features operated correctly. Feedback from the owner and staff of Donadel Hometel was also gathered to determine whether the system met the expected requirements and usability standards.
5. **Deployment and Implementation** – Once the system passed the testing procedures, the proposed system was deployed for initial implementation. The researchers provided demonstrations and orientation regarding the use of the system to ensure proper utilization by the intended users. The deployment phase included setting up the database, configuring the system environment, and preparing the system for operational use.
6. **Continuous Improvement** – The Agile methodology supports continuous improvement even after the initial implementation of the system. Feedback and observations gathered from users were used to identify possible enhancements and future improvements for the system. This phase ensures that the proposed system remains adaptable and capable of addressing the changing needs of the business.

3.3 System Architecture

The proposed system follows a web-based client-server architecture composed of three major components: the user interface, application logic, and database management system.

The user interface serves as the front-end component where customers and administrators interact with the system through a web browser. Customers can submit

reservation requests, while administrators can manage bookings, schedules, and reports.

The application logic acts as the processing component of the system. It handles reservation validation, booking updates, occupancy calculations, report generation, and calendar management processes.

The database component stores all reservation records, customer information, booking schedules, occupancy data, and generated reports. The database ensures organized data storage and efficient retrieval of reservation information.

The client-server architecture enables real-time processing of reservation data and supports centralized management of booking records.

3.4 Tools and Technologies Used

The development of the proposed system utilized various tools and technologies to ensure efficient system performance and user accessibility.

- **Frontend:** HTML, CSS, and JavaScript were used in developing the user interface of the system. These technologies were utilized to create responsive web pages, interactive elements, and user-friendly navigation.
- **Backend:** PHP was used as the server-side scripting language responsible for processing reservation transactions, managing user requests, and handling system functionalities.
- **Database:** MySQL was used as the database management system for storing reservation records, customer details, occupancy data, and generated reports.
- **Development Tools:** Visual Studio Code served as the integrated development environment (IDE) for coding, debugging, and managing the system development process.

3.5 Data Gathering Techniques

To gather the information needed for this study, different ways of collecting data were used to look into how reservations are managed and to identify problems in Doandel Hometel.

- **Surveys:** Structured survey questionnaires were distributed to the owner and staff of Donadel Hometel to identify common challenges encountered in reservation management. The survey also gathered information regarding user expectations and desired system features.
- **Interviews:** Interviews were conducted with the owner and selected personnel to obtain detailed information regarding current reservation procedures, booking issues, and operational concerns.
- **Observations:** Watching how things happen in real-time helped to record the actual reservation processes, including data entry, schedule management, and mistakes that happen. This way gives insight into the practical workflows that surveys or interviews might miss.

3.6 Testing Procedure

- **Unit Testing:** Each part of the system or its functions was checked on its own to make sure that every section, like adding, changing, or removing reservations, works correctly.
- **System Testing:** The whole system was looked at as one unit to guarantee that everything works well together, including the user interface, backend, and database.
- **User Acceptance Testing (UAT):** The owner and employees tried out the system using actual reservation information to see how well it works, how easy it is to use, and how it performs. The feedback they gathered pointed out areas where improvements could be made.

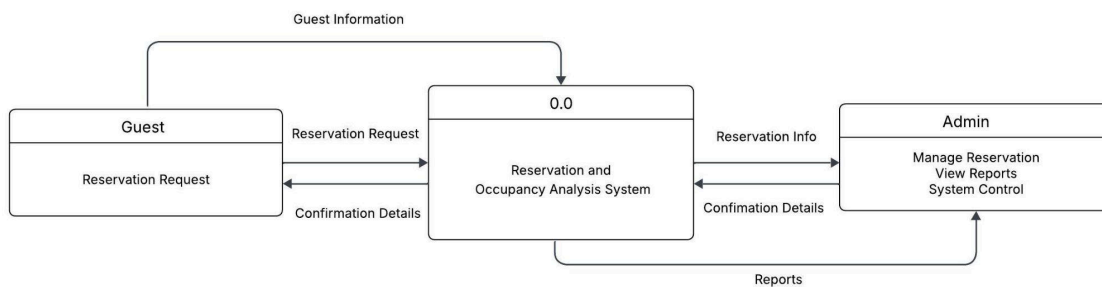
3.7 Evaluation Criteria

The system was evaluated based on these points:

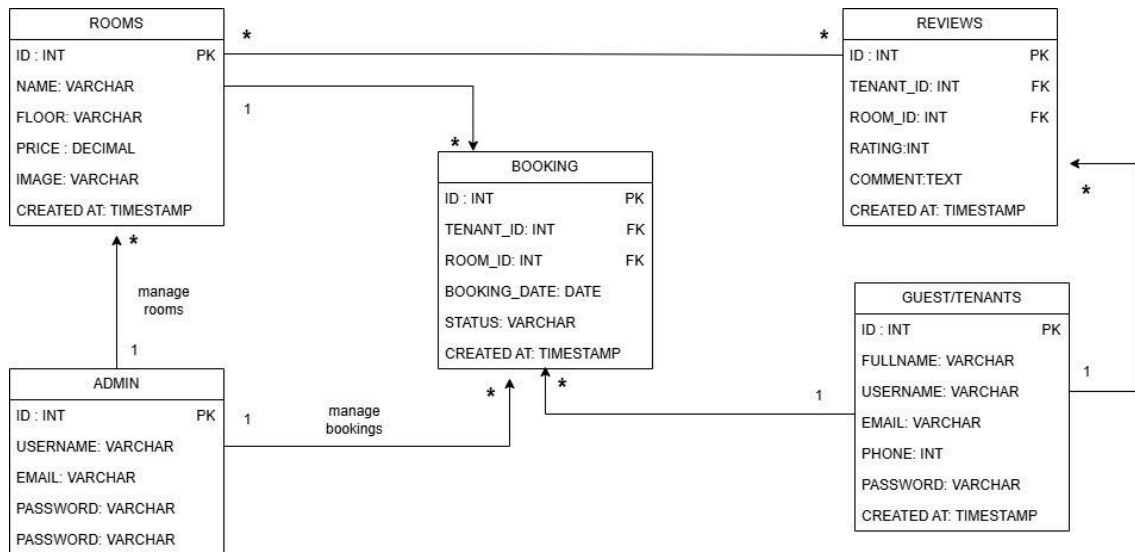
- **Functionality:** Checks how well the system can handle tasks related to managing bookings, like making, changing, and deleting records.
- **Usability:** Look at how easy it is for staff and administrators to use the system.
- **Efficiency:** Looks at how fast and accurately the system can manage bookings and retrieve information.
- **Reliability:** Examines how consistently the system works without making mistakes or stopping.
- **Accuracy:** Confirms how correct the reservation details are when they are saved and shown.

Supporting Diagrams

Data Flow Diagram (DFD – Level 0)



Entity Relationship Diagram



Use Case Diagram

